SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – SIH25048**
- **Problem Statement Title- Gamified Learning Platform for Rural Education**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID- 83063**
- **Team Name- QuantumCodersx**

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EduQuest - Gamified STEM Learning Platform for Rural Students

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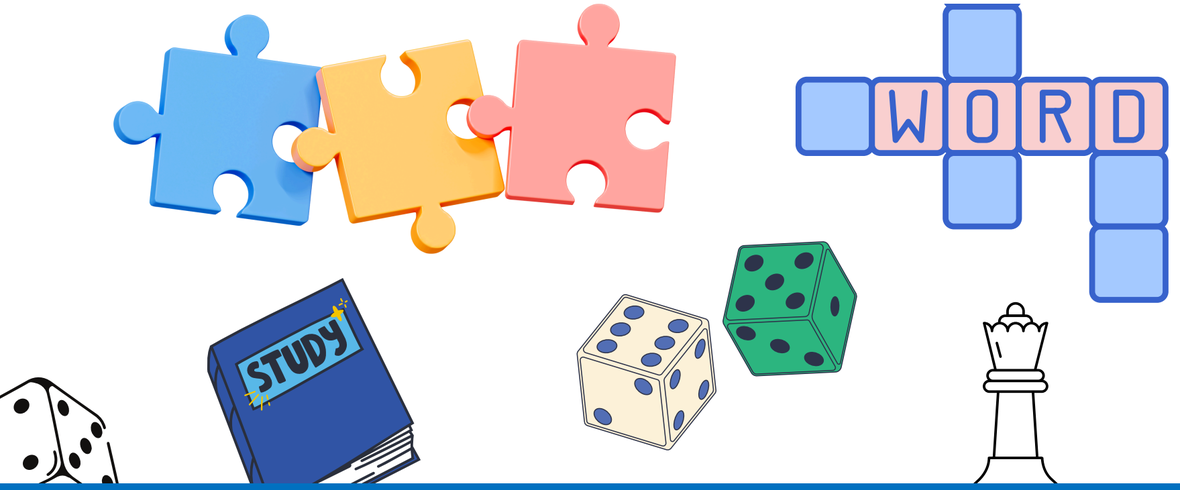
Proposed Solution

- *Develop an interactive gamified learning platform for rural students (Grades 6–12).*
- *Focus on STEM subjects through games, quizzes, puzzles, and simulations.*
- *Provide multilingual content to ensure inclusivity across regional languages.*
- *Enable offline access with data sync when internet is available.*

Addresses the Problem

- *Increases student engagement through **rewards, leaderboards, and challenges.***
- *Breaks **language barriers** by delivering localized educational content.*
- *Solves connectivity issues in rural areas with **offline features**.*
- *Supports **teachers dashboards** for monitoring progress.*



Uniqueness of the Solution – EduQuest

- ***Multilingual support** (English, Hindi, Odia) for inclusive learning.*
- ***Teacher–Student collaboration** hub for doubt solving, progress tracking, and badge assignment.*
- ***Offline-first approach** with data sync for rural areas with poor connectivity.*
- ***Gamified reward system** with coins, badges, and levels (Bronze, Silver, Gold).*

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TECHNICAL APPROACH

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SOFTWARES

FRONTEND
- React
- TypeScript
- Vite

BACKEND
- Node
- JavaScript

DATABASE
- SQLite3

Methodology and Process for Implementation EduQuest

EduQuest

- **Requirement Analysis** — Identify needs of rural students and teachers
- **System Design** — Design UI/UX, database, and gamification logic
- **Development** — Develop frontend, backend, and integrate content
- **Testing** — Conduct unit, integration, and usability testing
- **Deployment** — Deploy on low-cost devices and provide training
- **Monitoring & Improvement** — Track engagement and collect feedback for updates

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FEASIBILITY AND VIABILITY

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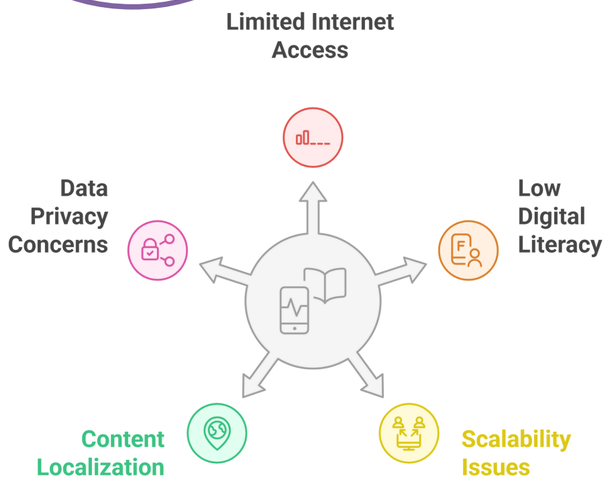

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- *Technically feasible using **open-source tools & lightweight frameworks.***
- *Platform works seamlessly on **low-cost devices (mobile, tablet, desktop).***
- ***Cost-effective** – requires only basic infrastructure for deployment.*
- ***Scalable design** – easy integration of new features & regional languages.*
- ***Operational feasibility** – user-friendly for both students & teachers.*

POTENTIAL CHALLENGES

- ***Limited internet access** in rural/remote areas.*
- ***Low digital literacy** among students and teachers.*
- ***Scalability issues** when handling large numbers of users.*
- ***Content localization** in multiple regional languages.*
- ***Data privacy & security** concerns in student/teacher profiles.*

STRATEGIES TO OVERCOME CHALLENGES

- ***Offline-first design** with auto sync when internet is available.*
- ***Capacity building** – training workshops for teachers & awareness drives.*
- ***Cloud-based infrastructure** to ensure scalability and performance.*
- ***Partnerships with local educators** for accurate translations.*
- ***Secure authentication** (OTP, encryption) for safe data handling.*
- ***Gamification & rewards** to maintain long-term engagement.*

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IMPACT AND BENEFITS

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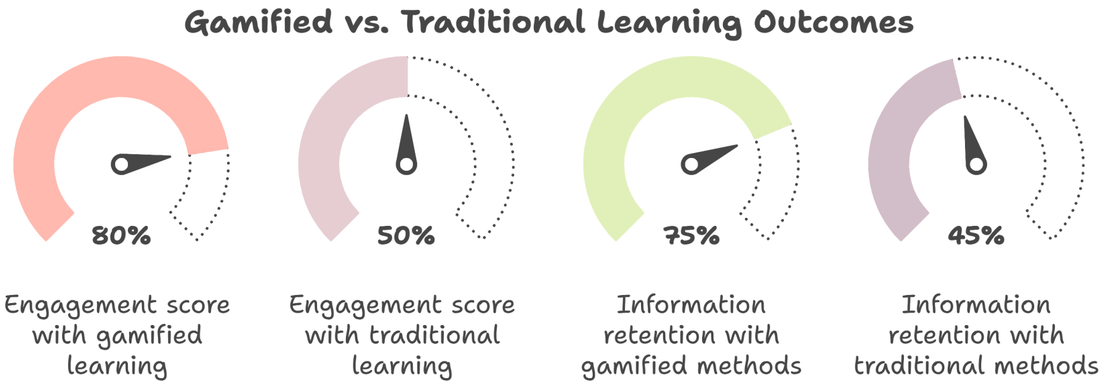
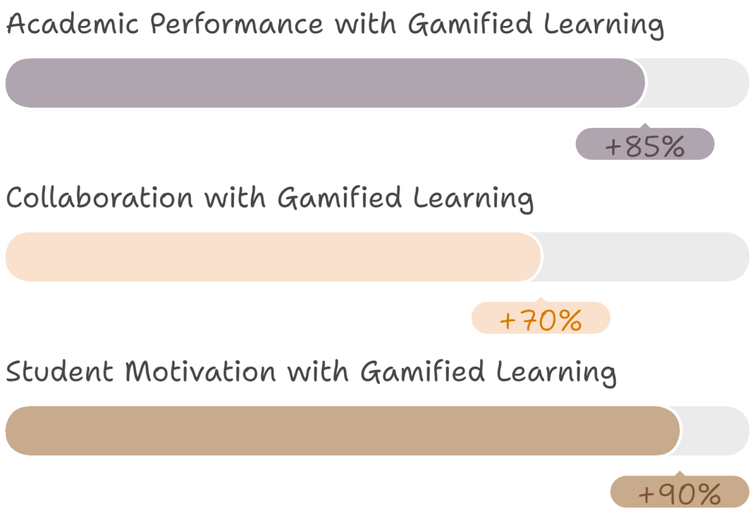
POTENTIAL IMPACT ON TARGET AUDIENCE

- *Increases STEM learning engagement among rural students (Grades 6–12).*
- *Bridges the digital divide by providing offline, gamified learning access.*
- *Empowers teachers with analytics to track student performance.*
- *Encourages multilingual learning, promoting inclusivity.*





BENEFITS OF THE SOLUTION

- ***Social***: *Improves quality of education in underserved areas, fosters equal opportunities.*
- ***Economic***: *Cost-effective, scalable on low-cost devices, reduces dependency on expensive solutions.*
- ***Environmental***: *Digital learning reduces paper usage and promotes eco-friendly education.*
- ***Long-term***: *Builds curiosity, problem-solving, and critical thinking skills among students, nurturing future innovators.*

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RESEARCH AND REFERENCES

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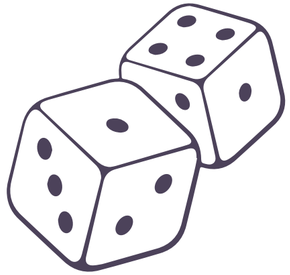
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  👉 *asercentre.org*
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